

Northstar Components

Forecast-Driven Next-Period Production Plan — Management Report

Warehouse NSC-01 · Currency EUR · Prepared 18 July 2026 · Orchestrated demand-forecast + optimization task

Executive Summary

This report determines the most profitable feasible integer production plan for Northstar Components' next period. Unlike a fixed demand plan, the demand ceilings here are estimated from each product's synthetic sales history and the known explanatory variables for the coming period, then fed into a resource-constrained optimization.

The workflow was orchestrated across two Optees capabilities: a linear regression (ml.regression.linear) was fitted separately for Product A and Product B to forecast next-period demand, and a mixed-integer linear program (milp.linear) then maximized total contribution subject to the forecast demand ceilings, minimum commitments, and hard resource capacities.

The recommended plan is to produce 18 units of Product A and 6 units of Product B, for a maximum total contribution of EUR 1,050. The optimization was mathematically optimal and independently verified as feasible. The forecasts themselves are point estimates and carry uncertainty; they should not be treated as certain, and the plan should be revisited if realised demand or the explanatory assumptions differ.

Workflow Overview

The analysis proceeded in four stages, each using a versioned Optees payload that was validated before execution:

- Stage 1 — Observed data: read unit economics and per-unit resource usage (Products), hard capacities (Resources), 12 months of historical demand with explanatory variables (History A, History B), and next-period explanatory variables (Next Period).
- Stage 2 — Forecasting: fit an ordinary-least-squares linear regression for each product (demand as a function of marketing spend, season index, and unit price), then apply the fitted model to the Next Period explanatory values to obtain a demand point forecast.
- Stage 3 — Optimization: solve a mixed-integer linear program that maximizes total contribution, using the demand forecasts as upper bounds, the product minimum commitments as lower bounds, and the three hard resource capacities as constraints.
- Stage 4 — Reporting: separate observed data, model predictions, and optimization decisions, and report mathematical status distinctly from independent validation.

Observed Workbook Data

The following values were read directly from the workbook and were not modified.

Product	Unit contribution (EUR)	Machine h/unit	Material kg/unit	Labour h/unit	Min units
A	40	2	3	1	5
B	55	4	2	2	3

Resource	Available capacity	Unit	Hard constraint
Machine time	60	hours	Yes
Raw material	80	kg	Yes
Direct labour	35	hours	Yes

Next-period explanatory variables (from the Next Period sheet; the demand forecast field was blank and marked 'To be estimated'):

Product	Marketing spend (kEUR)	Season index	Unit price (EUR)
A	4	1.2	10
B	7	1.2	15

Model Predictions (Demand Forecast)

A separate OLS linear regression was fitted for each product on its 12 historical observations, with demand explained by marketing spend, season index, and unit price. Both models reproduced the historical data essentially exactly ($R^2 = 1.00$ on both train and test partitions; residuals at floating-point noise level), indicating the synthetic history follows an exact linear rule. The recovered relationships are:

- Product A: demand = $20 + 3 \times (\text{marketing spend}) + 5 \times (\text{season index}) - 2 \times (\text{unit price})$
- Product B: demand = $10 + 2 \times (\text{marketing spend}) + 4 \times (\text{season index}) - 1 \times (\text{unit price})$

Applying each fitted model to the next-period explanatory variables gives the demand point forecasts:

Product	Forecast demand (units)	Used as integer demand cap	Model fit R^2
A	18.0	18	1.00
B	13.8	13	1.00

These are point estimates, not certainties. The perfect in-sample fit reflects a clean synthetic dataset and does not guarantee future accuracy; Optees explicitly warns that the metrics describe this train/test split only and do not establish causality or guarantee predictive performance. The forecasts were used as demand upper bounds; because production is in whole units, the effective integer ceilings are 18 for A and 13 for B.

Optimization Decisions — Recommended Quantities

The MILP maximized $40 \cdot A + 55 \cdot B$ over integer A, B, subject to A between 5 and 18, B between 3 and 13, and the three hard resource constraints. The optimal decision is:

Product	Description	Recommended units	Unit contribution (EUR)	Contribution (EUR)
A	Precision housing	18	40	720
B	Control module	6	55	330
Total		24		1,050

Expected Contribution

Expected total contribution is EUR 1,050 — EUR 720 from Product A ($18 \times \text{EUR } 40$) and EUR 330 from Product B ($6 \times \text{EUR } 55$). This figure is conditional on the demand forecasts being realised; if actual demand for A falls below 18 units, the achievable contribution would be lower.

Resource Utilization

Resource	Used	Available	Utilization	Status
Machine time (h)	60	60	100%	Binding
Raw material (kg)	66	80	82.5%	Slack: 14 kg
Direct labour (h)	30	35	85.7%	Slack: 5 h

Usage evaluated at $A=18$, $B=6$ using the per-unit coefficients: machine $2A+4B=60$, material $3A+2B=66$, labour $A+2B=30$.

Binding Constraints

- Machine time (60 hours) is fully consumed and is the sole binding resource — the plant bottleneck.
- Product A forecast demand cap (18 units) is binding — A is produced up to its full forecast ceiling.
- Product B is interior: at 6 units it sits above its minimum (3) and below its forecast ceiling (13), constrained instead by the machine-time trade-off (each B consumes 4 machine hours versus 2 for A).
- Raw material and direct labour are non-binding, with 14 kg and 5 hours of slack.

Uncertainty

The plan rests on forecasts, so its economics are uncertain in ways the optimizer cannot capture:

- Demand forecasts are point estimates. Although in-sample fit is perfect, real demand can deviate from any model, and the forecast should be read as a central estimate rather than a guaranteed quantity.
- The regression was trained on only 12 synthetic observations per product with a fixed 10/2 train/test split; small samples make out-of-sample behaviour hard to guarantee.
- The next-period explanatory variables (marketing spend, season index, price) are themselves planning assumptions; if they change, the forecast and therefore the optimal plan change.
- Because Product A is produced exactly at its forecast ceiling, any downward error in A's demand directly reduces realised contribution; the plan has no demand buffer on A.

Assumptions

- Next-period demand equals the fitted OLS prediction at the given explanatory variables; forecasts are used as demand upper bounds, rounded down to whole units for feasibility ($A \leq 18$, $B \leq 13$).

- Minimum commitments ($A \geq 5$, $B \geq 3$) and hard capacities (machine 60 h, material 80 kg, labour 35 h) are taken from the workbook and treated as hard constraints.
- Unit contributions and per-unit resource coefficients are taken directly from the Products sheet; production is integer-valued.
- Administrative Notes fields (warehouse code, currency, colour, energy review) are contextual and were not converted into constraints. The energy review supplies no numeric limit, so no energy constraint was modelled.
- No value was silently invented: the blank Next Period demand fields were estimated by the regression, not assumed.

Limitations

- Data are fully synthetic; figures are illustrative and not from a real company.
- The regression capability has no independent Optees validator, so its output carries model-fit metrics but no external feasibility certificate; only the optimization stage is independently validated.
- The model is a single-period, deterministic plan and does not handle inventory carryover, setup times, price/demand elasticity beyond the fitted linear form, or stochastic demand.
- The MILP validator confirms feasibility, integrality, and objective consistency but does not by itself prove optimality nor assess whether the model captures business intent.

Mathematical Status and Independent Validation

Mathematical status (solver-reported) and independent validation (external checks) are reported separately below for each Optees stage.

Stage / Capability	Mathematical status	Independent validation
Forecast A — ml.regression.linear	Trained / feasible (OLS, $R^2=1.00$)	Not available (no validator registered)
Forecast B — ml.regression.linear	Trained / feasible (OLS, $R^2=1.00$)	Not available (no validator registered)
Optimization — milp.linear	OPTIMAL (CBC, relative gap 0)	VERIFIED — 5/5 checks passed, 0 violations

Every payload was validated against its versioned schema before execution (all valid, no warnings). For the optimization, the solver reports the solution as provably optimal, and the Optees validator independently confirms it is feasible (bounds, integrality, all three constraints satisfied) and that the reported objective of EUR 1,050 equals the recomputed objective. For the two forecasting stages, Optees reports no independent validator is registered, so those results are supported by model-fit metrics only — not an external validation certificate.